

3.1 Define management information system.

An MIS is a system that provides managers with the necessary information to make decisions about an organization's operations. The MIS gathers data from various sources and processes it to provide information tailored to the managers' and their staffs needs.

While businesses use different types of systems, they all share one common goal: to provide managers with the information to make better decisions. In today's fast-paced business environment, having access to accurate and timely information is critical for success. MIS allows managers to track performance indicators, identify trends, and make informed decisions about where to allocate resources.

Importance of management information systems for businesses

MISs allows businesses to have access to accurate data and powerful analytical tools to identify problems and opportunities quickly and make decisions accordingly. A management information system should do the following:

- ❖ Provide you with information you need to make decisions
- ❖ Can give you a competitive edge by providing timely, accurate information
- ❖ Can help you improve operational efficiency and productivity
- ❖ Allows you to keep track of customer activity and preferences
- ❖ Enables you to develop targeted marketing campaigns and improve customer service

3.2 Explain the structure of decision making in organizations e.g. structured, semi-structured etc.

What is an organizational structure?

An organizational structure details how certain activities are delegated toward achieving an organization's goal. It outlines an employee's role and various responsibilities within a company. The more authority employees have, the higher up they'll be on the organizational structure. In addition, the more organized a structure is, the more efficiently a company operates.

Types of organizational structures

There are four types of organizational structures. Understanding how they work and what their benefits and drawbacks are can help you make a more informed decision as to which to implement in your workplace. The four types are:

1. Functional structure: In a functional structure, organizations are divided into specialized groups with specific roles and duties. A functional structure is also known as a bureaucratic organizational structure and is commonly found in small to medium-sized businesses. Most people in the workforce have experience working in this type of organizational structure. For example, many companies divide their organization into various departments such as finance, marketing and human resources. Each of these departments then has a manager who oversees it. This manager is then supervised by an administrator or executive who oversees multiple departments.

Advantages of this structure may include:

- ❖ Employees grouped by skill
- ❖ Greater sense of teamwork

Disadvantages of this structure may include:

- ❖ Lack of communication with other departments
- ❖ Harmful competition
- ❖ Management issues

2. Divisional structure: In a divisional structure, various teams work alongside each other toward a single, common goal. Each of these divisions has an executive who manages how that branch operates, controls its budgets and allocates its resources. Large companies employ this type of organizational structure. One example of the divisional structure is a car company that separates its company into SUV, electric or sedan vehicle branches. While each branch has its own function, they all work toward the same goal of making a sale. This is also known as the multi-divisional structure.

Advantages of this structure may include:

- ❖ Focus on a single good or service
- ❖ More centralized leadership

Disadvantages of this structure may include:

- ❖ Poor integration with other divisions
- ❖ Competition between divisions
- ❖ Lack of communication between divisions
- ❖ Potential tax implications

3. Flatarchy: In a flatarchy, there are little to no levels of management. A company using this structure could have only one manager in between its executive and all other employees. It is called a flatarchy because it is a hybrid of a hierarchy and a flat organization. This type of organizational structure is used more by smaller companies since they have fewer employees, though it can be used in companies of all sizes. While some companies grow out of this organizational structure, others continue to use it.

Advantages of this structure may include:

- ❖ Cost-efficient
- ❖ Fosters good communication
- ❖ Higher employee morale
- ❖ Faster decision making

Disadvantages of this structure may include:

- ❖ Potential employee conflict

- ❖ Leadership confusion

4. Matrix structure: In the matrix-style of organizational structure, employees are divided into teams that report to two managers-a project or product manager along with a functional manager. In essence, a matrix structure is a combination of various organizational structures. Because these teams have two managers, a matrix structure promotes duality and the sharing of resources. Employees working for companies using the matrix structure have the potential to widen their skill set since they might be assigned to various projects requiring different levels of expertise or skills.

Advantages of this structure may include:

- ❖ Fosters open dialogue
- ❖ Flexible workplace environment

Disadvantages of this structure may include:

- ❖ Leadership confusion
- ❖ Conflicting leadership loyalties
- ❖ Potentially more costly
- ❖ Roles may not be clearly defined
- ❖ Potentially heavy employee-workload

3.3 Explain information flow.

There are three main types of information flow:

- **Internal information flow:** This is the exchange of information between different departments or units within a company. It helps to ensure that everyone is working towards the same goal.
- **External information flow:** This is the exchange of information between a company and its stakeholders, such as customers, suppliers, or partners. It helps businesses to build relationships and understand the needs of their stakeholders.
- **Inter-organizational information flow:** This is the exchange of information between two or more companies. It helps businesses to collaborate and compete in the marketplace.

3.4 Identify relationship between MIS and TPS.

TPS becomes a major source of data for MIS. The data that is generated through TPS is on the level of operations, such as payroll or order processing. TPS tracks daily routine transactions that are essential to conduct business. MIS makes heavy use of data from TPS though it also utilizes data from other sources.

3.5 Present information.

Information can be presented in multisensory ways and presentation can be adapted to individual needs, supporting independence.

A combination of visual and auditory stimuli is provided.

Students can learn at their own pace and revisit to consolidate learning as often as they like.

Differentiated work can be presented flexibly in order to meet individual learning requirements.

1. **Print:** Print media is a means of mass communication that allows people to access physical copies of your message and retain easy access to it. Some examples of print include newspapers, magazines and brochures.
2. **Leaflets:** A leaflet is a small sheet of printed paper used to convey a short message clearly and concisely. It combines explanatory texts with supporting graphics and aims to portray information in non-complex terms. Leaflets contain a minimum amount of detail to not overcomplicate the message.
3. **Electronically:** Presenting information electronically is becoming the default method of communication, as it allows you to engage with audiences no matter the distance. Some electronic means include email, social media, pay-per-click ads or website banners and other instant messaging systems.
4. **Videos:** Videos allow you to share educational, instructive or meaningful information in a way that's easy to retain. You may film yourself presenting the information or use images, captions or a voiceover instead. To present information via video is suitable if your messages rely on other video footage for meaning.
5. **Essays:** An essay is an extended piece of writing that outlines an argument and follows an established structure. An introduction outlines the argument being made, while the following paragraphs expand on the argument. Essays are suitable if you're attempting to explain a certain topic or standpoint.
6. **Speeches:** A speech is a way to present information that works well when you have a large audience. The quality of a speech depends on your confidence and presentation skills, which provide you with the ability to deliver information in a compelling and transformative way.
7. **Face-to-face:** Face-to-face is different from speeches or other in-person interactions, as it involves working with someone one on one. In-person meetings make it easier to assess the listener's reaction to information and facilitate better understanding where necessary.
8. **Presentations:** Presentations are similar to speeches in that they involve providing demonstrations or information to a large audience. Yet presentations often combine speech with images and short amounts of text to increase meaning.
9. **Visual aids:** Visual aids include graphs, photographs, video clips and other visual material that individuals use in addition to spoken information. They help reduce the monotony associated with speaking for long periods while holding notes. For instance, you can use graphs and diagrams to represent statistics or data you're discussing, allowing recipients to better visualize your message.

10. Info graphics: Info graphics are a collection of data visualizations, imagery and text used to present information quickly and clearly. They're useful when the information you want to convey is primarily visual. They help convey small amounts of information in an easily comprehensible way and as quickly as possible.

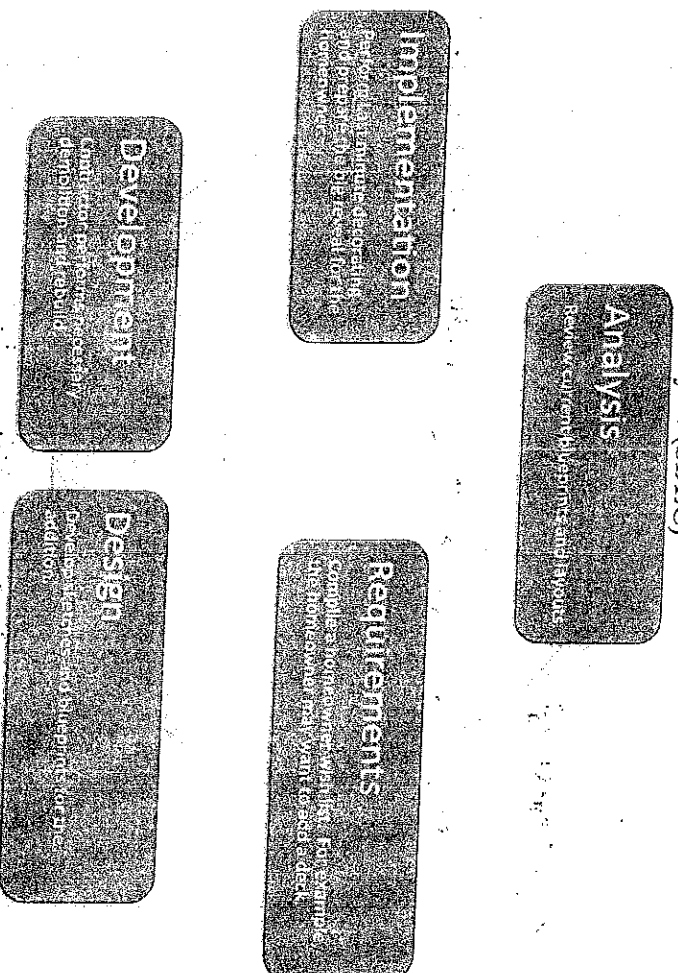
11. Articles: Articles are a form of written communication that intends to portray a central theme or idea to a readership. Similar to essays, they're long-form and intended to cover a certain topic. You can use articles to convey a definite point of view, describe various ideas, introduce a topic or write something purely descriptive.

12. Teleconference: A teleconference involves the live exchange of information remotely to people linked via a telecommunications system. This works if you're struggling to get everyone in the same room or working with dispersed teams across multiple locations. Teleconferences provide all the benefits of face-to-face interactions while allowing companies to save on travel costs and avoid scheduling problems.

13. Reports: A report is a written piece that uses facts to discuss issues, events, situations or findings. Writing a report allows you to present an argument in a formal and structured way. Reports are usually more evidence and research-based when compared to articles or essays.

3.6 Design view of information system.

Systems Development Life Cycle (SDLC)



Information systems are designed using the systems development life cycle (SDLC). The SDLC is to a large extent common sense spelled out in stages. First, analyze the current situation. Then specify the requirements that a solution should embody. The next stage is to design a solution (no programming yet). Then the system is developed (programmed) and tested. Finally, the system goes live for the end users as it is implemented in the business setting. To review, the five phases are:

1. Analysis
2. Requirements (vision of future state)
3. Design
4. Development
5. Implementation

3.7 Report information system.

An information reporting system: is a system that is used to process, store and retrieve information. They are mainly used in various industries including the healthcare industry for storing patient data in electronic medical records.

Different Types of Reporting

There are different kinds of reporting including:

- ❖ **Internal reporting:** in this type of system, employees report to their managers. Reports sent would be private and not shared with the public.
- ❖ **Public reporting:** in this type of system, reports are publicly available to anyone who searches for them on the internet (this includes companies that use Public Company Accounting Oversight Board or PCAOB).
- ❖ **Self-reporting:** in this type of system, reports are NOT publicly available and it is up to the company to decide whether or not these reports can be made available.
- ❖ **Internal auditing:** this type of system is used in many companies and it is considered to be the most effective of the four types. The company performs an internal audit on itself to ensure that it is following rules and regulations set out by the government or other private sector bodies but this still leaves room for bias and corruption because employees might not report all irregularities.